

ENERGY REQUIREMENTS DURING REPRODUCTION IN FEMALE COMMON SPINY MICE (*ACOMYS CAHIRINUS*)

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We hypothesized that energy intake of lactating females is the main factor determining growth of the young and that use of maternal energy reserves is less important when food is available in abundance. We studied reproducing female common spiny mouse (*Acomys cahirinus*) offered millet seeds and water ad libitum. Females gave birth to 1 or 2 young, and growth rate was similar between litters of 1 infant and 2 infants combined. In addition, energy intake of the 2 groups of females was similar. Lactating females increased energy intake by 45% (per g^{0.75}) over that of nonreproducing females. Energy intake and change in body mass of the lactating female each explained about 50% of the variance in infant growth and, therefore, our hypothesis was not supported. Energy intake and change in body mass of the lactating female together explained approximately 76% of the variance in growth of young.

Key words: *Acomys cahirinus*, body energy, energetics, lactation, metabolizable energy intake, pregnancy, reproduction

Reproduction is one of the most costly activities in terms of energy expenditure in female mammals. Consequently, caloric intake, in particular for small mammals, plays a major role in reproduction and, indeed, a number of studies have concluded that food availability seems to be the main factor determining reproductive activities and success (Bergallo and Magnusson 1999; Doonan and Slade 1995).

Small mammals usually consume enough food to provide energy for the support and maintenance of the pregnant female and for fetal growth and development. In fact, many females can consume food in excess of maintenance requirements for pregnancy and store energy that could be used during lactation. This has been demonstrated in the Djungarian hamster (*Phodopus sungorus*—Weiner 1987) and the cotton rat (*Sigmodon hispidus*—Randolph et al. 1995). Females

that do not consume enough energy generally use body energy reserves to maintain near-normal fetal development (Degen 1997).

Lactation is energetically much more demanding than gestation (Gittleman and Thompson 1988; Millar 1977). It is considered to be the most energy-expensive phase of reproduction for the female mammal and, as a consequence, is generally accompanied by a great increase in energy intake. Small lactating homeothermic mammals can often meet the additional energy demands by increasing food consumption (Glazier 1985; Kaczmarski 1966; Migula 1969; Poppitt et al. 1994; Rogowitz and McClure 1995); however, many are forced to use body energy reserves during lactation (Degen 1997; Kam and Degen 1993). For this, some rodents deposit reserves after conception that allows them to buffer or supplement food consumption (Degen 1997).

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Therefore, energy provided from food plus from mobilization of body tissue by the lactating female affects milk production, and as a result, offspring growth. With food available in abundance, we hypothesize that metabolizable energy intake is the main factor determining growth of young and that use of energy reserves is less important. To test this hypothesis, we measured energy intake of the nonpregnant, pregnant, lactating, and postlactating common spiny mouse (*Acomys cahirinus*) when offered millet seeds and water ad lib. We also determined the change in body mass of the female and took this change as an indication of change in body energy content. In addition, we determined the growth rate of the offspring. We then determined the effect of metabolizable energy intake and body mass change of the lactating female on the growth rate of her offspring.

MATERIALS AND METHODS

Study species and site of capture.—The common spiny mouse (*Acomys cahirinus*; Muridae—average adult body mass about 45 g) is widely distributed among rocky habitats from eastern and northern Africa to Iran (Harrison and Bates 1991; Mendelssohn and Yom-Tov 1987). It is common in Israeli deserts where it inhabits natural crevices among rocks; it does not dig burrows. This omnivorous rodent has a varied diet consisting of moist and dry vegetation, seeds, and animal material (Degen et al. 1986; Happold 1984). It must be supplied with water when consuming only seeds (Daily and Haines 1981; Shkolnik and Borut 1969).

Reproduction is spread out throughout the year but occurs mainly from the end of winter throughout spring till late summer (March–September). Gestation is 37–40 days and lactation lasts about 21 days. Females can reproduce 2–3 times per year and litter size is 1–4 young. Young are precocial and are born with well-developed senses. Their bodies are covered with short hair and they are able to move around by the 2nd day (Mendelssohn and Yom-Tov 1987). *A. cahirinus* was trapped in the Negev highlands near Sde Boqer (30°52'N, 34°47'E; 475 m elev.). The region is considered an arid zone having mild winters and warm summers with 250–300

dry days/year and with a ratio of precipitation to evaporation of <0.2 (UNESCO 1977). Mean annual precipitation is 91 mm, all occurring in winter, with large variation among years.

Animal maintenance and measurements.—The study was conducted at a mean air temperature of 25°C and a photoperiod of 12L:12D. Eighteen adult *A. cahirinus* females were used and were offered millet seeds and drinking water ad lib. throughout the study. The study was divided into 4 periods, during which time digestibility trials were conducted to determine metabolizable energy intake: before pregnancy for 14 days, during pregnancy for 21 days starting 7–10 days after the onset of pregnancy, during lactation for 21 days, and postweaning for 7 days.

For digestibility trials before and during pregnancy and postweaning, females were maintained in individual metabolic cages (11 by 13 by 20 cm). During lactation, the females and young were maintained in breeding cages (13 by 24 by 34 cm). Both types of cages allowed for measurement of food intake and fecal output. After the measurements before pregnancy, females were kept together with several males for 5 days and were then placed in their metabolic cages. If pregnant, they were kept there until end of the weaning period and, if not pregnant, were placed with males again. Of the 18 females, 15 became pregnant. We had planned to determine dry-matter intake of young by collecting and weighing their feces as described by Kam and Degen (1993). However, as less than 1.5% of total feces was from the young, we considered their dry-matter intake as negligible.

To determine dry-matter content of seeds, daily samples were weighed and oven-dried at 100°C to constant mass. These samples, as well as daily feed remains and feces, were measured for caloric value using a ballistic bomb calorimeter (model CBB-370, Gallenlamp, Loughborough, Leicestershire, United Kingdom) with benzoic acid ($26,453 \pm 3.9$ J/g; BCS-CRM No. 190n, Bureau of Analysed Samples Ltd., Bristol, United Kingdom) as a standard. Dry-matter intake (g) was calculated as the difference between dry matter offered (g) and dry matter not consumed (g), and dry-matter digestibility (% of dry-matter intake) was calculated from the difference between dry-matter intake (g) and dry-matter fecal output (g). Digestible energy intake (kJ) was calculated as the difference between gross energy intake (kJ) and fecal energy output

(kJ), and metabolizable energy intake (kJ) was taken as 0.98 of digestible energy intake (Grodzinski and Wunder 1975). Digestible energy (%) was calculated as the percentage of digestible energy intake of gross energy intake.

Females were weighed (to 0.01 g, Model 500 PJ, Precisa, Milton Keynes, United Kingdom) daily before food was offered. To calculate daily change in body mass during lactation (loss in body mass), we regressed natural logarithm of percentage of body mass over time for each animal using body mass of 1st day as 100% (Kleiber 1975). During lactation, offspring were weighed (to 0.01 g) daily and, using regression analysis, rate of daily growth was determined.

Statistical analysis.—Regression analyses, with both single and multiple independent variables, were done using Statistica for Windows software (StatSoft, Inc., Tulsa, Oklahoma), and $P < 0.05$ was chosen as the minimal acceptable level of significance. The effects of metabolizable energy intake and of change in body mass of the lactating female on growth rate of offspring were determined. Metabolizable energy intake was expressed in 3 ways: as total intake; as relative increase (intake during lactation divided by intake before pregnancy); and as increase above nonpregnant state (intake during lactation minus intake before pregnancy). Time-series analysis was used to test whether there was a trend in metabolizable energy intake with time in any of the periods (Kendall 1976). Analysis of covariance (ANCOVA) was used to assess differences in metabolizable energy intake among periods, taking body mass and litter size as covariate factors. Thus, effects of body mass and litter size on metabolizable energy intake were removed. Values are presented as means $\pm$ SD.

RESULTS

Time-series analysis did not show a trend in metabolizable energy intake with time in any of the periods and, therefore, mean value for each animal in each period was used in analyses; that is, we used 1 value per animal per period. ANCOVA showed that metabolizable energy intake differed among all periods ($F = 25.56$, $d.f. = 3, 54$, $P < 0.01$).

Gross energy of seeds was 17.6 ± 0.29 kJ/g dry matter. Measurements on dry-mat-

ter digestibility and energy digestibility did not differ among the 4 periods—dry-matter digestibility was $89.4 \pm 2.1\%$, digestible energy, as percentage of gross energy, was $88.5 \pm 1.4\%$, and metabolizable energy was $86.5 \pm 1.3\%$.

Four females gave birth to 1 young each and 11 females gave birth to litters of 2. Initial body mass per young was 6.74 ± 1.30 g for litters of 1 and 6.67 ± 1.40 g for litters of 2. During lactation, growth rate of singletons, 0.66 ± 0.23 g/day, was the same, statistically, as combined growth rate for litters of two, 0.54 ± 0.33 g/day. Therefore, in analyses using change in body mass of offspring, change for a single offspring and for twins combined were used.

Females gained body mass during pregnancy but lost body mass during lactation. Values for metabolizable energy intake of females that gave birth to singles or twins were similar in all periods of measurement and were analyzed together. Before pregnancy, metabolizable energy intake (BPMEI) of females was 436.9 ± 43.5 kJ $\text{kg}^{-0.75} \text{ day}^{-1}$. During pregnancy, metabolizable energy intake increased by 9.6% and during lactation (LMEI) by 45.2%. Metabolizable energy intake during the 1st week after weaning was 19.2% higher than that before pregnancy, although body masses during these periods were similar (Table 1).

Effect of net increase in metabolizable energy intake during lactation (LMEI – BPMEI) on the growth rate of the pup (Fig. 1) could be described as Δm_b (g/day) = $0.1764 + 0.0149(\text{LMEI} - \text{BPMEI})$ (kJ/day), with $n = 15$, $s_{yx} = 0.200$, $r^2 = 0.51$, $P < 0.01$. When expressed in absolute terms or as the multiple of BPMEI, LMEI had similar effects on growth rate of young, with similar r^2 (0.48 and 0.47), s_{yx} (0.206 and 0.210), and P values ($P < 0.01$ and $P < 0.01$).

Effect of daily change in body mass of the lactating female (ΔM_b) on daily change in body mass of the young (Δm_b) took the form (Fig. 2) Δm_b (g/day) = $+0.1587 -$

TABLE 1.—Body mass and metabolizable energy intake of *Acomys cahirinus* females before and during pregnancy, during lactation when nursing 1 or 2 young, and postweaning.

Stage	Time period (day)	Body mass (g)		Metabolizable energy intake ($\text{kJ kg}^{-0.75} \text{ day}^{-1}$)	
		Mean	SD	Mean	SD
Before pregnancy	14	49.4	7.0	436.9	43.5
During pregnancy	21 ^a	65.8	6.3 ^b	478.7	60.1
During lactation	21				
Initial		56.2	8.0		
Final		46.1	9.9	634.2	109.6
Postweaning	7	47.6	5.9	521.0	71.4

^a For 21 days of the 37–40-day pregnancy (see text).

^b Mean mass during pregnancy.

$0.5486\Delta m_b$ (g/day), with $n = 15$, $s_{yx} = 0.210$, $r^2 = 0.46$, $P < 0.01$.

The combined effect of these 2 independent variables on daily change of body mass of the offspring took the form Δm_b (g/day) = $-0.0274 + 0.0133(\text{LMEI} - \text{BPMEI})$ (kJ/day) + $0.4171\Delta m_b$ (g/day) with $n = 15$, $s_{yx} = 0.135$, $r^2 = 0.76$, $P < 0.01$.

DISCUSSION

Energy intake increased only slightly during pregnancy, as has been observed in a number of rodent species (Degen 1997). However, energy cost of lactation was much higher. Metabolizable energy intake by the lactating female can be partitioned into energy required for maintenance and that required for milk production. When offered food ad lib., some small lactating females such as the pine vole (*Microtus pinetorum*—Lochmiller et al. 1982) and golden-mantled ground squirrel (*Spermophilus saturatus*—Kenagy et al. 1989) remain approximately in energy balance. Others such as the lesser hedgehog tenrec (*Echinops telfairi*—Poppitt et al. 1994) can add to their energy reserves, whereas still others such as the Djungarian hamster cannot assimilate enough energy and must mobilize energy reserves (Weiner 1987). The common spiny

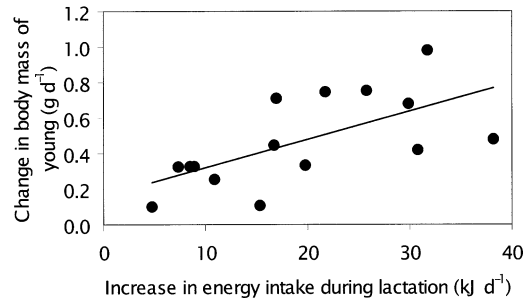


FIG. 1.—Effect of net increase in metabolizable energy intake during 21 days of lactation (LMEI – BPMEI) on growth rate (Δm_b) of *Acomys cahirinus* young. Line was fitted by linear regression, taking the form Δm_b (g/day) = $0.1764 + 0.0149(\text{LMEI} - \text{BPMEI})$ (in kJ/day); $s_{yx} = 0.200$, $r^2 = 0.51$, $P < 0.01$.

mouse fell into this last group and mobilized energy reserves.

During pregnancy, embryos grew from 0.0 to 6.7 g per infant and maternal tissue increased by 6.8 g (Table 1). To calculate increase in energy content, we used energy equivalents of 2.76 and 10.9 kJ/g for embryo mass and maternal mass change, respectively (Kam and Degen 1993). Consequently, energy content per infant at birth was 18.5 kJ and increase in energy of maternal tissue was 74.0 kJ for a total increase of 92.5 kJ for a female with 1 infant and 111 kJ for a female with 2 infants. During pregnancy, females increased metabolizable

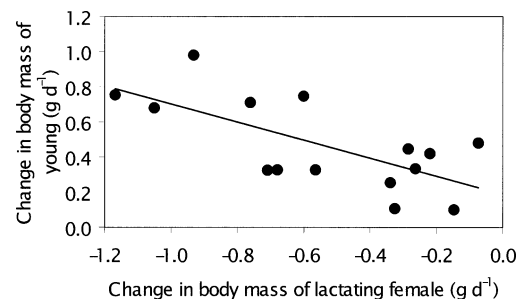


FIG. 2.—Effect of change in body mass (Δm_b) of lactating *Acomys cahirinus* female during 21 days of lactation on growth rate of young (Δm_b). Line was fitted by linear regression, taking the form Δm_b (g/day) = $0.1587 + 0.5486\Delta m_b$ (g/day); $s_{yx} = 0.210$, $r^2 = 0.46$, $P < 0.01$.

energy intake by $41.8 \text{ kJ kg}^{-0.75} \text{ day}^{-1}$ or 200.9 kJ in total. Thus, increase in energy retention of embryos and female is approximately 50% of increase in metabolizable energy intake. The difference could be because of heat increment of feeding for growth, energy content of the placenta, and an increase in energy needs of the pregnant female (Degen 1997).

Lactating common spiny mice increased their energy intake by 45% per $\text{g}^{0.75}$ over that of the nonpregnant female. This is somewhat lower than that found for most rodents. Increase in nutritional energy intake by the lactating female above that of nonpregnant female rodents usually ranges between 65% and 200% (Gittleman and Thompson 1988; Glazier 1985; Kaczmarski 1966; Mattingly and McClure 1982; Migula 1969; Millar 1978; Poppitt et al. 1994; Randolph et al. 1977). However, in small rodents and other small mammals, this increase may be even higher and can reach 400% in the house mouse, *Mus musculus* (König and Markl 1987), and even 800% in a small shrew, *Sorex coronatus* (Genoud and Vogel 1990).

During 21 days of lactation, young from litters of 1 and 2 grew from 6.7 to 20.6 g and maternal tissue decreased by 10.1 g (Table 1). To calculate change in energy content, we used energy equivalents of 5.29 and 10.9 kJ/g for change in infant mass and maternal mass, respectively (Kam and Degen 1993). Consequently, increase in body energy of the litter from birth to 21 days was 73.4 kJ and decrease in energy of maternal tissue was 110.1 kJ. During the 21 days of lactation, females increased metabolizable energy intake by $197.3 \text{ kJ kg}^{-0.75} \text{ day}^{-1}$ or 446.0 kJ in total. Therefore, energy increase above maintenance for females during lactation was 556.1 kJ ($446.0 + 110.1$). Consequently, growth in body energy of the litter is approximately 13% of the increase in energy above maintenance for the females. The difference could be because of milk production and increase in energy needs of the lactating female. Fur-

thermore, there is an added heat increment of feeding for milk production and for mobilizing maternal tissue. In young, milk intake is used for maintenance, growth (energy retention), and heat increment of feeding for growth (Degen 1997; Kam and Degen 1993).

Equivocal results have been reported regarding effect of litter size on energy intake in lactating females. Lactating fat sand rats (*Psammomys obesus*) increase their metabolizable energy intake over maintenance requirements of nonreproducing females by up to 85% when nursing 1–3 young, by up to 118% when nursing 4–5 young, and by up to 138% when nursing 6–8 young (Kam and Degen 1993). Similarly, an increase in energy intake with an increase in litter size has been reported in the pine vole (*Microtus pinetorum*—Lochmiller et al. 1982) and the white-footed mouse (*Peromyscus leucopus*—Millar 1978). However, no difference in energy intake was found in lactating common spiny mice females with different litter sizes. Females nursing 1 young increased metabolizable energy intake by 46%, while females nursing 2 young increased metabolizable energy intake by 44%. This similar increase in metabolizable energy intake in the 2 groups could explain, at least in part, why growth rate of a single infant was similar to the combined rates of 2 infants.

Body mass of females after weaning was not significantly different than before pregnancy. Interestingly, metabolizable energy intake in the week after weaning was 19% higher than before pregnancy. Part of this was most likely because of a carryover of high intake during lactation.

In this study, spiny mice had litters of 1 or 2 young; however, this species typically has litters of 1–4 (Mendelssohn and Yom-Tov 1987). Perhaps the diet of only millet seeds had an effect on fertility, because this species is an omnivore consuming green vegetation and animal matter in addition to seeds (Degen et al. 1986). Larger litter sizes suggest that females would require greater

metabolizable energy intake, and this might be possible with a more varied diet. Otherwise, females would have to use more of their energy reserves and/or young would have to consume more food. The precocial nature of the young would be amenable to this.

Metabolizable energy intake, expressed as total intake, intake relative to nonproducing females, or net intake above that of nonproducing females, explained approximately 50% of the variance in change in body mass of growing young. Standard error of the estimate was similar for all methods. Change in body mass of the lactating female similarly explained about 50% of the variance in infant growth. Therefore, the 2 independent factors (metabolizable energy intake and energy mobilized from body tissue) were of similar importance in explaining variance in the rate of infant growth and, consequently, our hypothesis was not supported. Metabolizable energy intake and change in body mass of lactating females together explained about 76% of the change in body mass of growing young.

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